

Report: Evolutionary Dynamics and Speciation under Environmental Change

1. Introduction

This study examines how the rate of environmental change influences speciation in a population evolving under Fisher’s Geometric Model. The central question is:

How does the rate of environmental change (parameter c) affect speciation?

We hypothesize that higher values of c increase selective pressure and phenotypic divergence, leading to more clusters and potentially faster adaptation.

2. Methodology

2.1 Model Overview

We simulate an evolutionary system with:

- Asexual reproduction
- Isotropic mutation
- Two-stage selection
- A dynamically shifting environment controlled by c

Individuals are represented as phenotype vectors in a continuous space, and fitness depends on distance to the environmental optimum.

2.2 Environmental Dynamics

The environment evolves as:

$$\alpha(t) = \alpha(t-1) + c + \eta$$

where:

- c is the deterministic environmental shift,
- $\eta \sim \mathcal{N}(0, \delta^2 I)$ is stochastic noise.

2.3 Fitness Function

Fitness is defined as:

$$\phi(p, \alpha) = \exp\left(-\frac{|p - \alpha|^2}{2\sigma^2}\right)$$

where p is the phenotype, α the optimum, and σ controls selection strength.

2.4 Mutation

Mutation is isotropic:

- Occurs with probability μ
- Each trait mutates with probability μ_c
- Mutation effect:

$$\Delta p_i \sim \mathcal{N}(0, \xi^2)$$

2.5 Selection

Selection consists of:

1. Threshold filtering: individuals with fitness below θ are removed
2. Wright–Fisher sampling proportional to fitness

2.6 Reproduction

Reproduction is asexual:

- Offspring are copies of parents with mutations
 - No recombination
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3. Experimental Design

3.1 Parameter Setup

Environmental change rates:

$$c \in 0.0, 0.002, 0.005, 0.01$$

Other fixed parameters:

- $N = 100, n = 5, T = 200$
- $\mu = 0.005, \mu_c = 0.1, \xi = 0.05$
- $\sigma = 0.5, \theta = 0.2$

3.2 Replication

Each configuration was run 20 times with different random seeds.

3.3 Metrics

Recorded per generation:

- Mean fitness
- Distance to optimum

- Phenotypic variance
- Number of clusters
- Silhouette score

Clusters serve as a proxy for speciation.

3.4 Speciation Definition

Speciation occurs when:

- $n_{\text{clusters}} \geq 2$
 - silhouette > 0.9
 - Conditions persist for above 30 generations
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4. Results

4.1 Mean Fitness

- Stable and high for $c = 0$
- Decreases as c increases

Higher environmental change reduces adaptation ability.

4.2 Lag from Optimum

- Low c : population tracks optimum
- High c : lag increases over time

Faster environmental change reduces tracking efficiency.

4.3 Phenotypic Variance

- Variance decreases over time in all cases
- Little difference between c values

Diversity loss is mainly driven by selection.

4.4 Speciation

- Most likely at intermediate c
- Low c : insufficient divergence
- High c : unstable dynamics prevent clustering

Speciation is maximized at moderate environmental change.

4.5 Speciation Time

- Mostly early events
- Some delayed cases

- Multimodal distribution

4.6 Correlations

- c negatively correlates with speciation
- σ positively correlates with speciation
- Mutation parameters have weaker effects

4.7 Interactions

- Highest speciation probability at:
 - Moderate c
 - High σ

Speciation depends on nonlinear interactions between parameters.

5. Discussion

Results show a nonlinear relationship between environmental change and speciation:

- Low $c \rightarrow$ weak divergence
- Moderate $c \rightarrow$ optimal speciation
- High $c \rightarrow$ unstable adaptation prevents clustering

Selection strength plays a key role, while mutation modulates variability.

Speciation emerges from interactions between c , σ , and mutation parameters rather than any single factor.